

(12) **United States Patent**
Hong et al.

(10) **Patent No.:** **US 12,406,880 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **DIELECTRIC SILICON NITRIDE BARRIER DEPOSITION PROCESS FOR IMPROVED METAL LEAKAGE AND ADHESION**

23/5226; H01L 23/53238; H01L 21/76883; H01L 21/76826; H01L 21/02312; H01L 21/02211; H01L 21/76834; H01L 23/53295

(71) Applicant: **Texas Instruments Incorporated**,
Dallas, TX (US)

See application file for complete search history.

(72) Inventors: **Qi-Zhong Hong**, Richardson, TX (US);
Joseph Jian Song, Plano, TX (US);
Gregory Boyd Shinn, Dallas, TX (US);
Bhaskar Srinivasan, Allen, TX (US)

(56) **References Cited**

U.S. PATENT DOCUMENTS

(73) Assignee: **TEXAS INSTRUMENTS INCORPORATED**, Dallas, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 409 days.

7,309,651 B2	12/2007	West et al.	
2006/0027924 A1 *	2/2006	Chen	H01L 21/288
			257/E21.174
2006/0154471 A1 *	7/2006	Minami	H01L 21/76807
			438/618
2006/0166491 A1 *	7/2006	Ida	H01L 21/76801
			257/E21.579
2007/0037388 A1 *	2/2007	Hohage	C23C 16/0227
			257/E21.224
2012/0205814 A1 *	8/2012	Wu	H01L 21/02203
			438/653
2013/0221535 A1 *	8/2013	Ma	H01L 23/53295
			428/408
2017/0125450 A1 *	5/2017	Hodo	H10D 86/0231

(Continued)

(21) Appl. No.: **17/751,976**

(22) Filed: **May 24, 2022**

(65) **Prior Publication Data**

US 2023/0386907 A1 Nov. 30, 2023

Primary Examiner — Douglas W Owens

(74) Attorney, Agent, or Firm — Xiaotun Qiu; Frank D. Cimino

(51) **Int. Cl.**

H01L 21/768 (2006.01)

H01L 21/02 (2006.01)

H01L 23/522 (2006.01)

H01L 23/532 (2006.01)

(52) **U.S. Cl.**

CPC **H01L 21/76829** (2013.01); **H01L 21/0217** (2013.01); **H01L 21/02274** (2013.01); **H01L 21/76877** (2013.01); **H01L 23/5226** (2013.01); **H01L 23/53238** (2013.01)

(58) **Field of Classification Search**

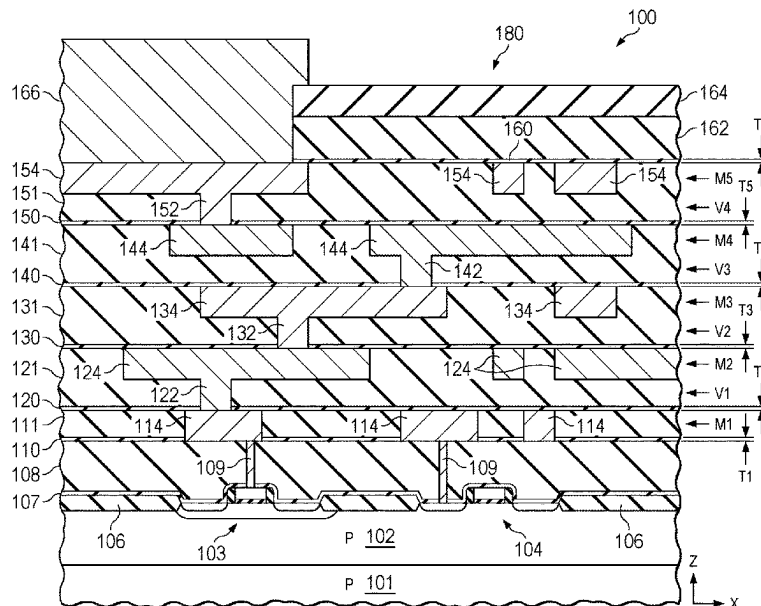
CPC H01L 21/76829; H01L 21/0217; H01L 21/02274; H01L 21/76877; H01L

(57)

ABSTRACT

An electronic device includes a semiconductor die having a multilevel metallization structure including stacked levels with respective dielectric layers and metal lines, and a low leakage, low hydrogen diffusion barrier layer on one of the stacked levels. The diffusion barrier layer contacts a side of the dielectric layer and the metal line of the one of the stacked levels, and the diffusion barrier layer includes silicon nitride material having a first bond percentage ratio of ammonia to silicon nitride that is greater than a second bond percentage ratio of silicon hydride to silicon nitride.

20 Claims, 19 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2020/0066987	A1 *	2/2020	Sims	C23C 16/4554
2021/0257293	A1 *	8/2021	Lee	H01L 21/76829
2023/0121958	A1 *	4/2023	Lee	H01L 21/02178
				257/758

* cited by examiner

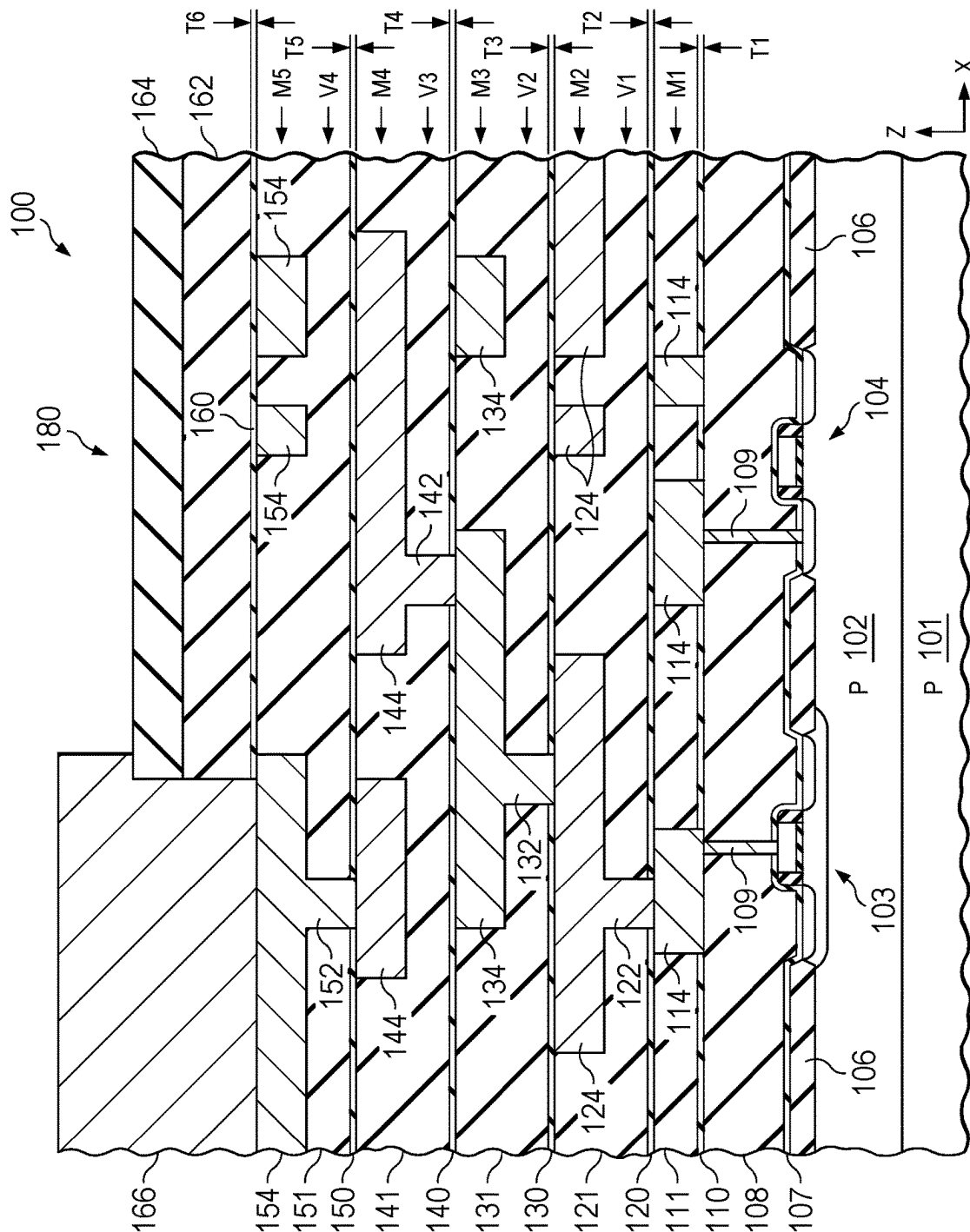


FIG. 1

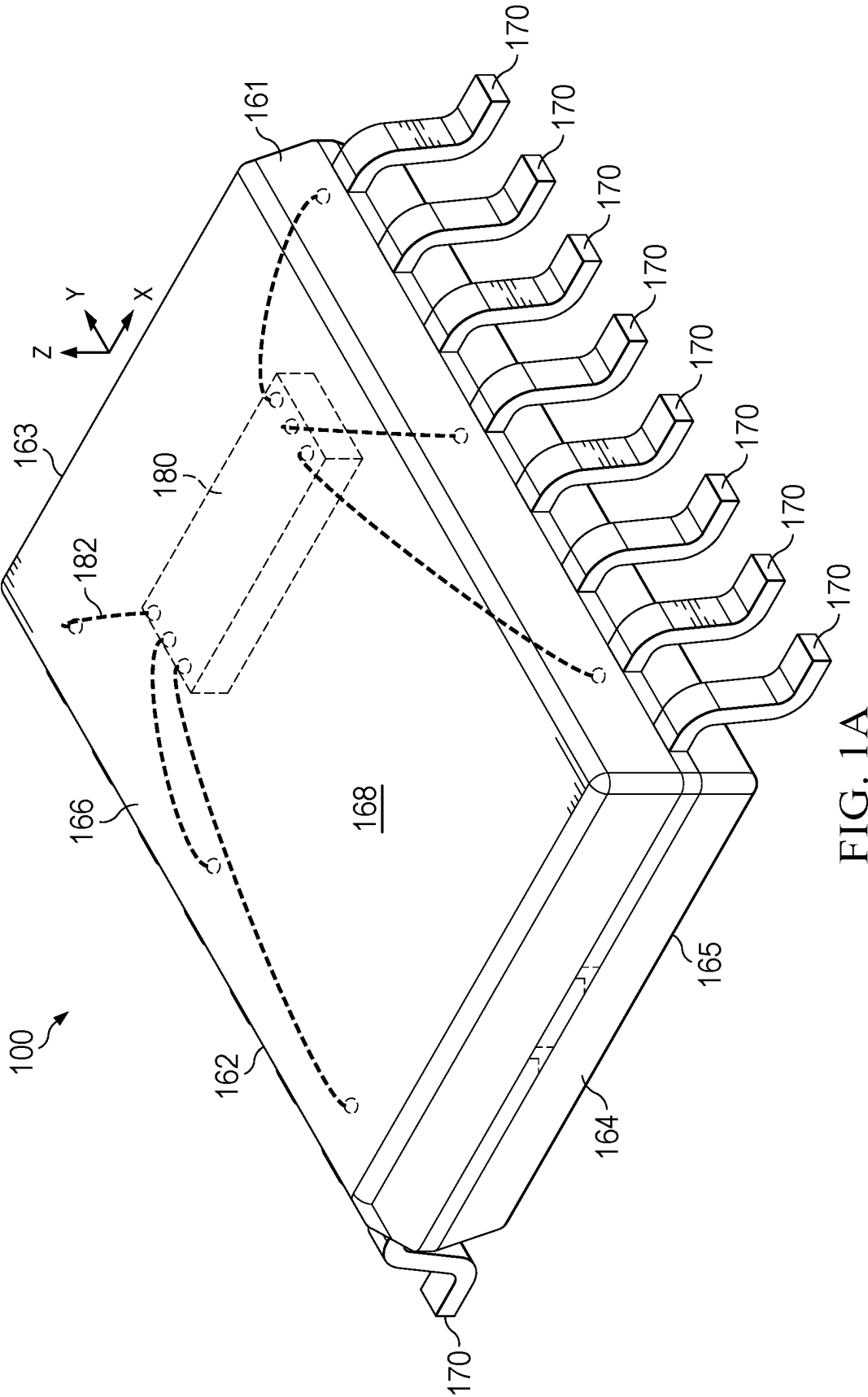


FIG. 1A

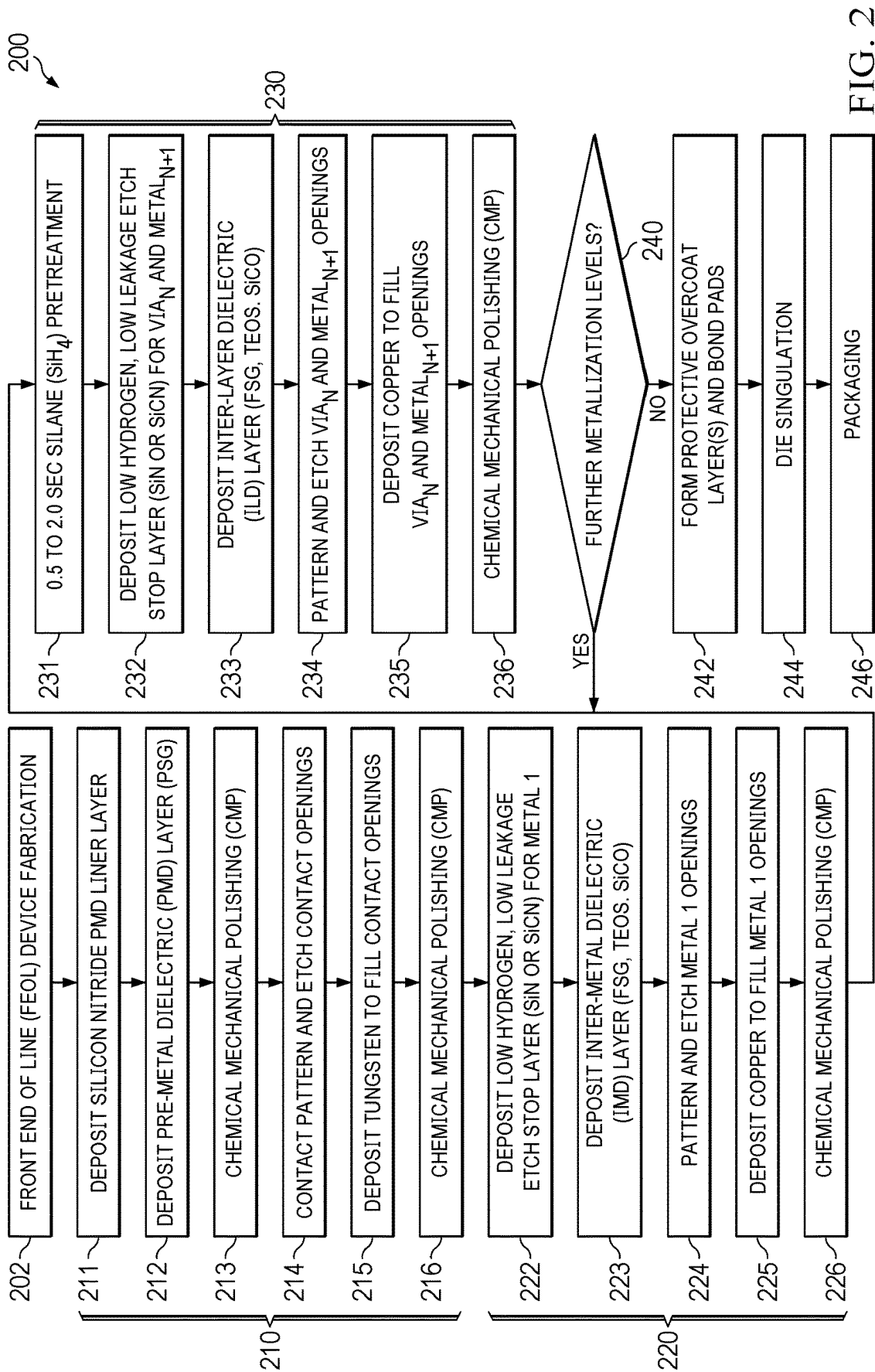


FIG. 2

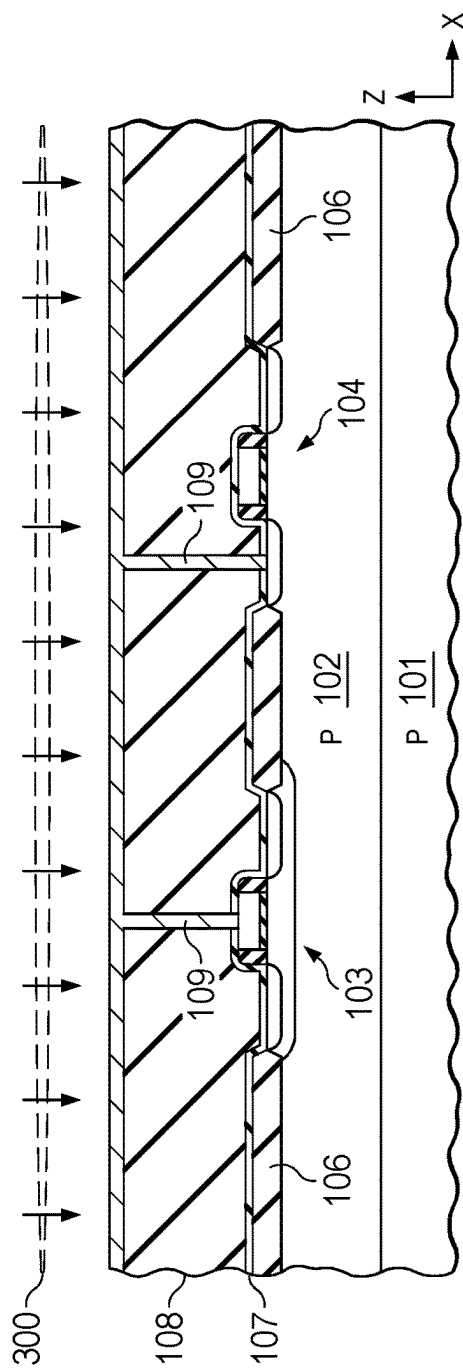


FIG. 3

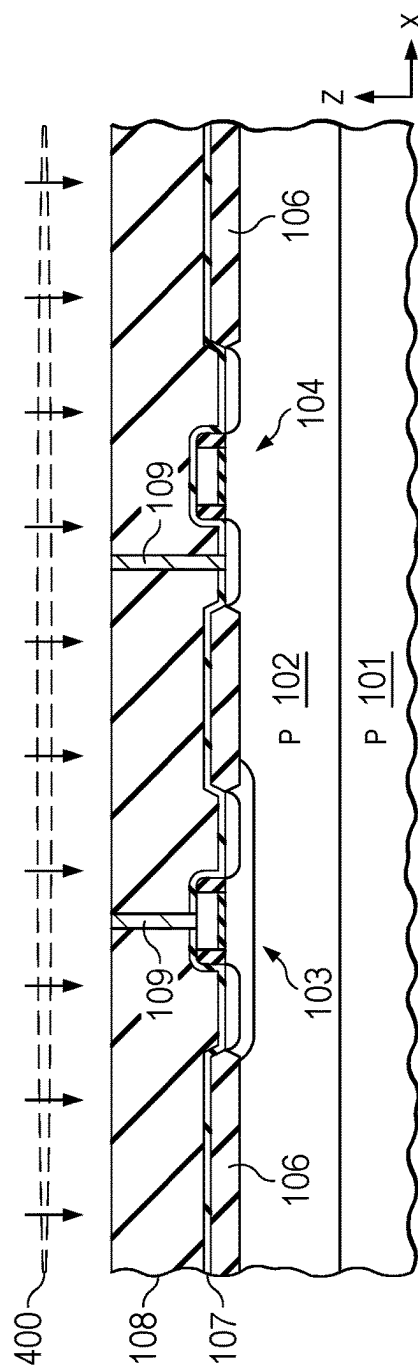


FIG. 4

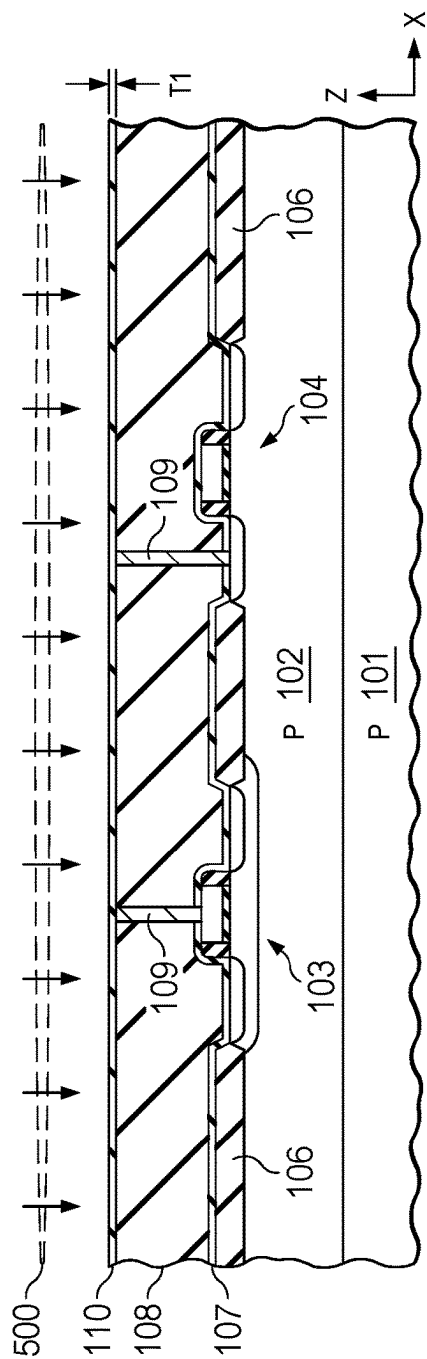


FIG. 5

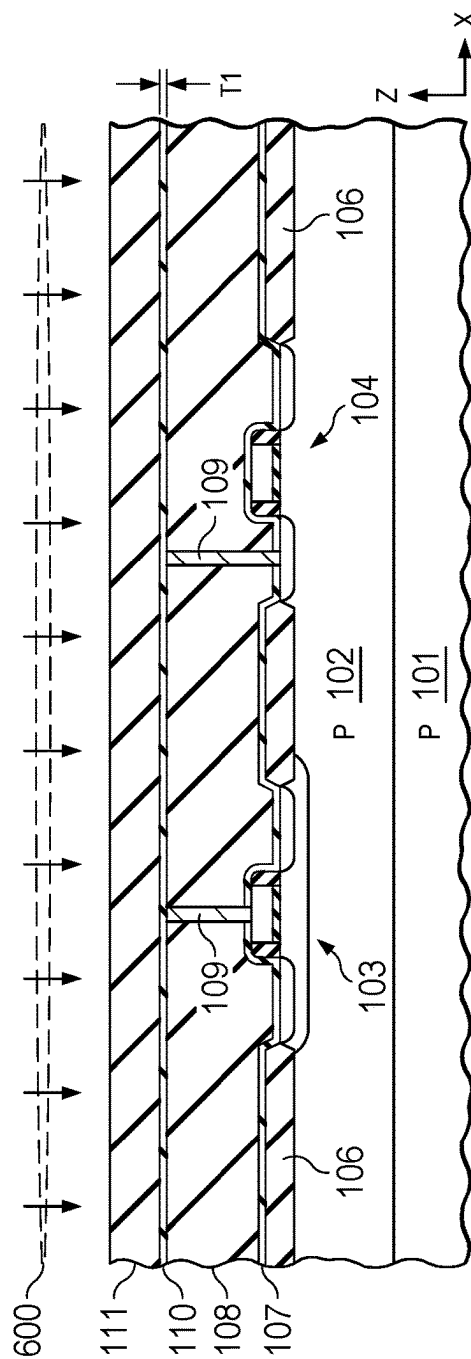


FIG. 6

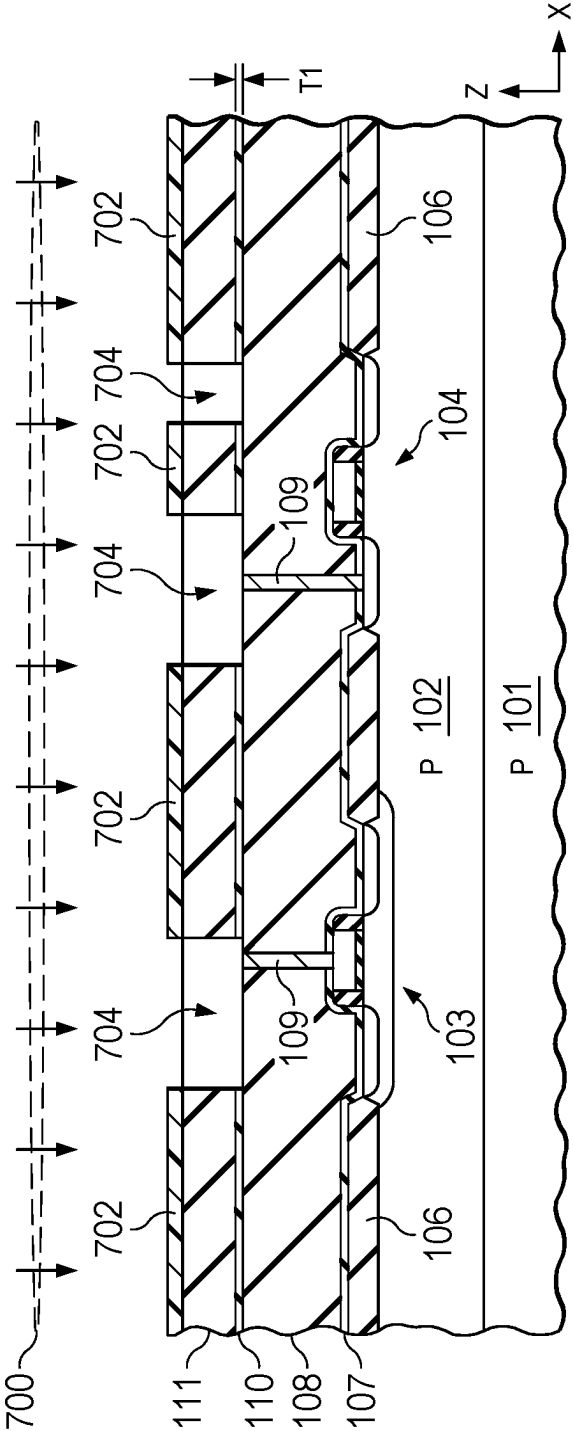


FIG. 7

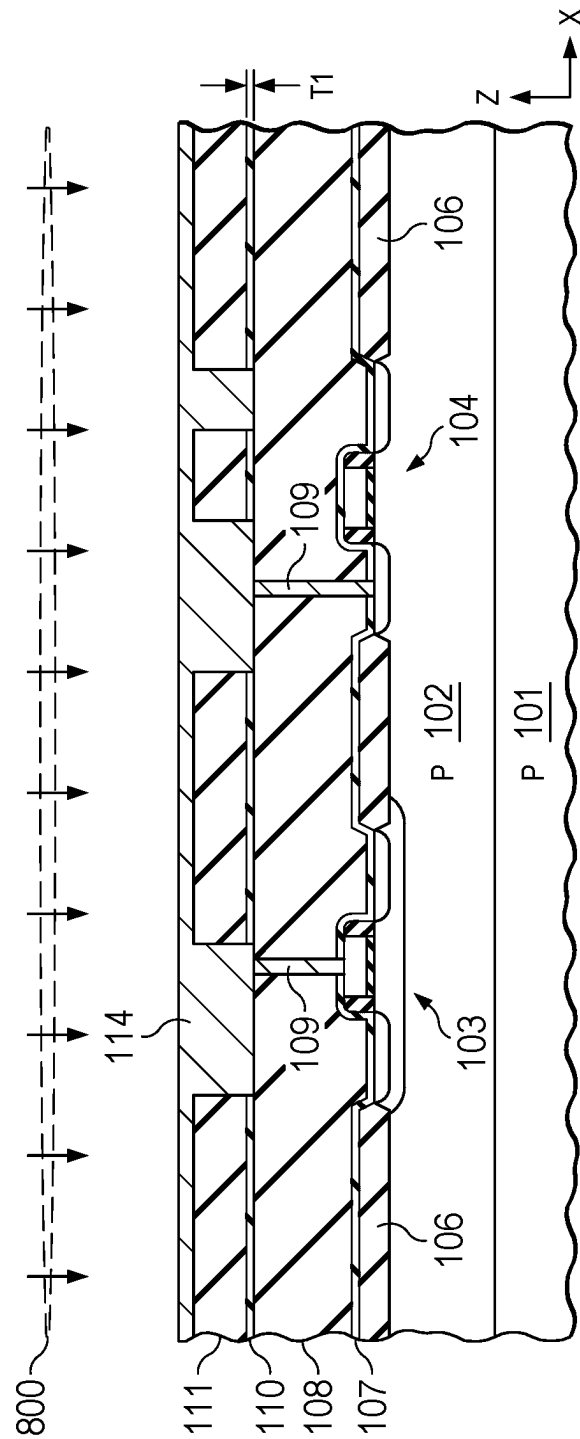


FIG. 8

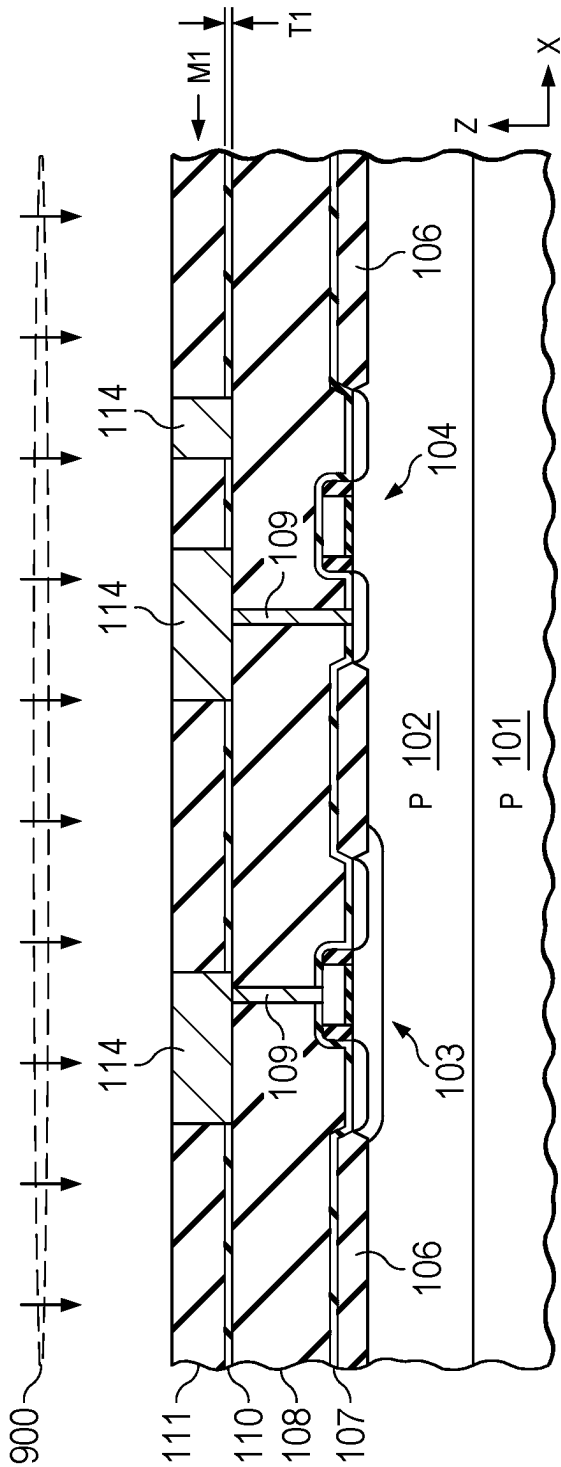


FIG. 9

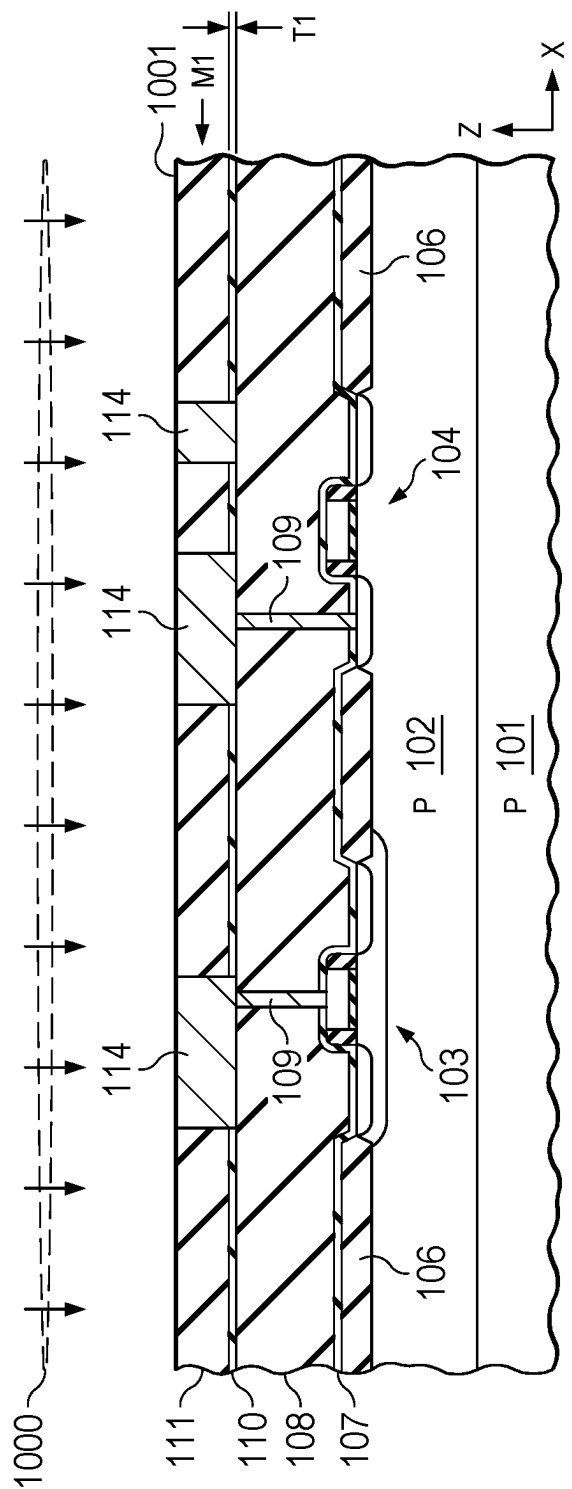


FIG. 10

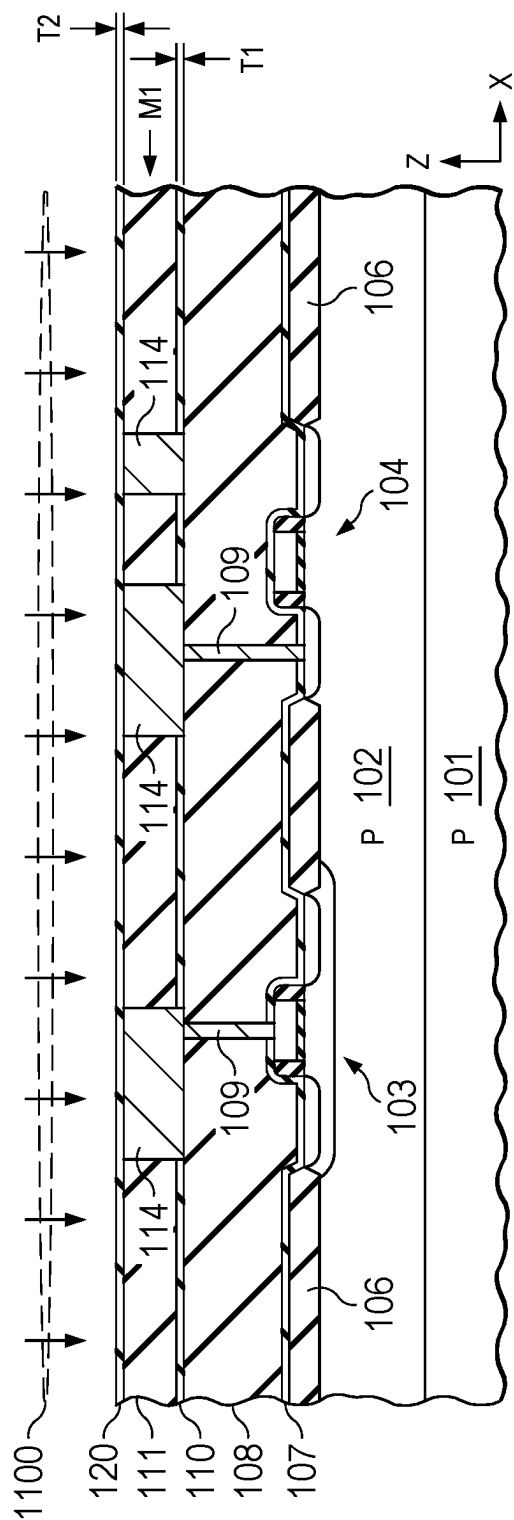


FIG. 11

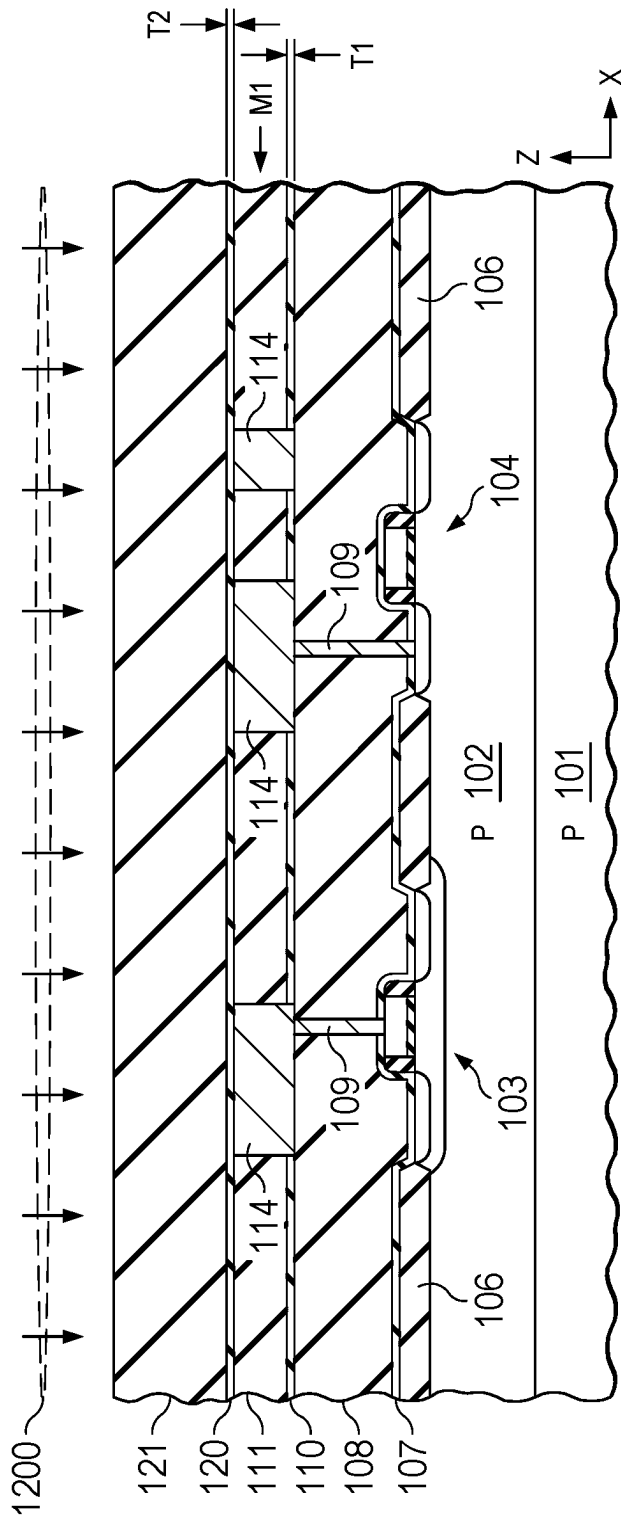


FIG. 12

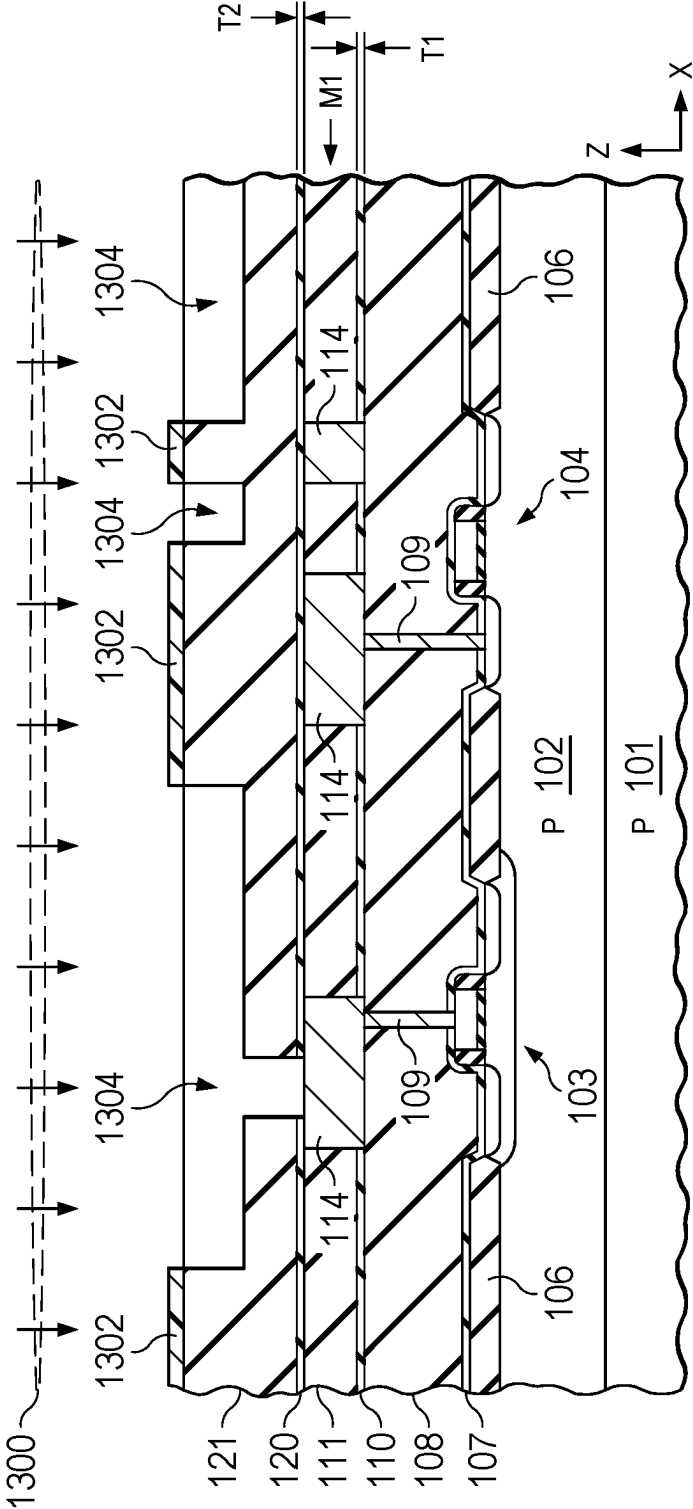


FIG. 13

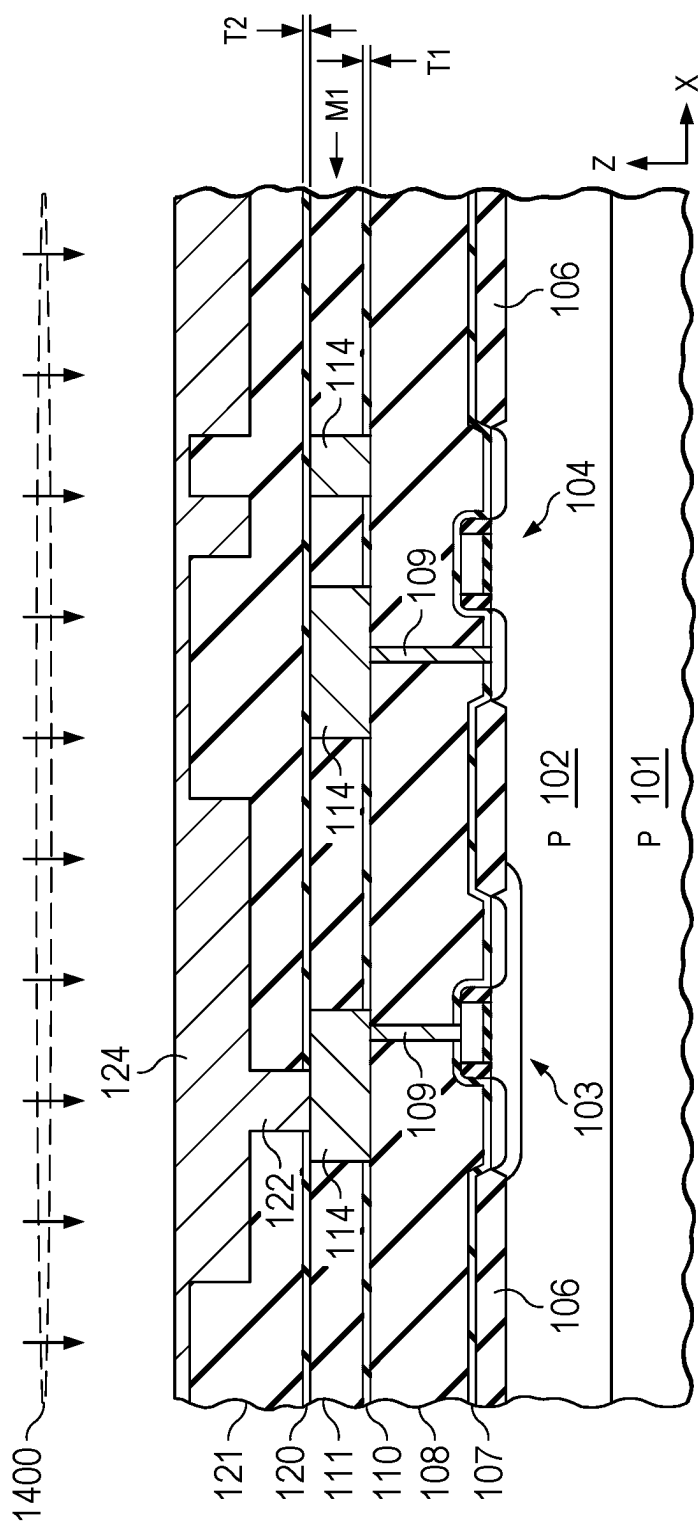


FIG. 14

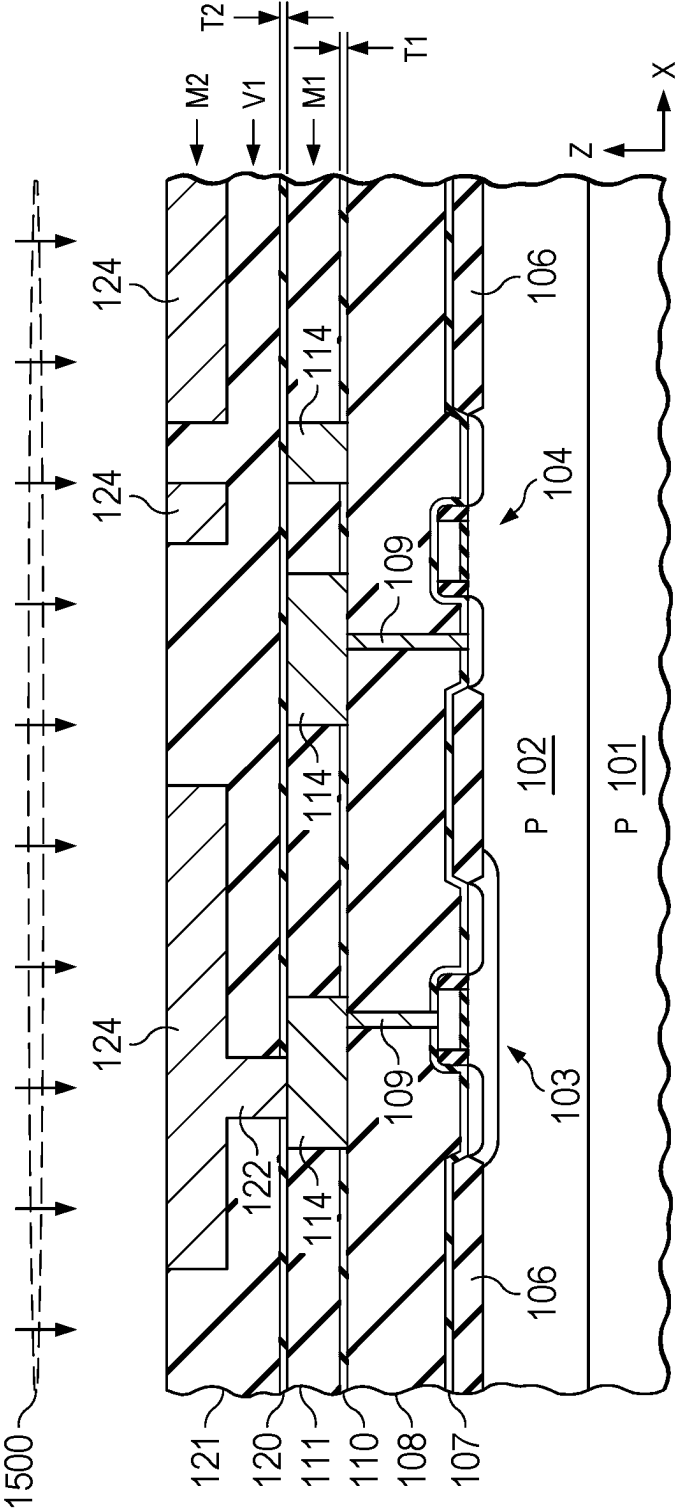


FIG. 15

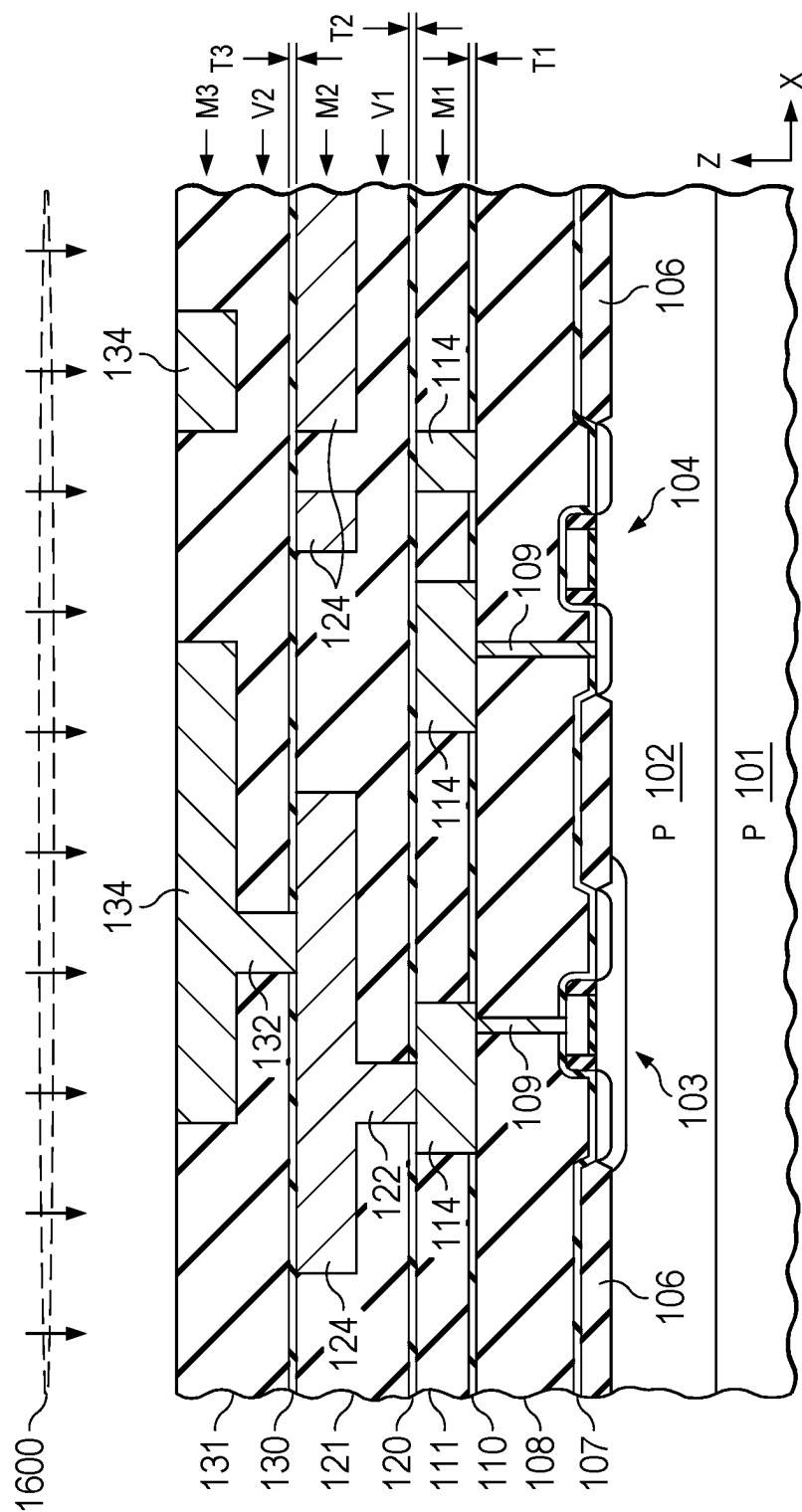


FIG. 16

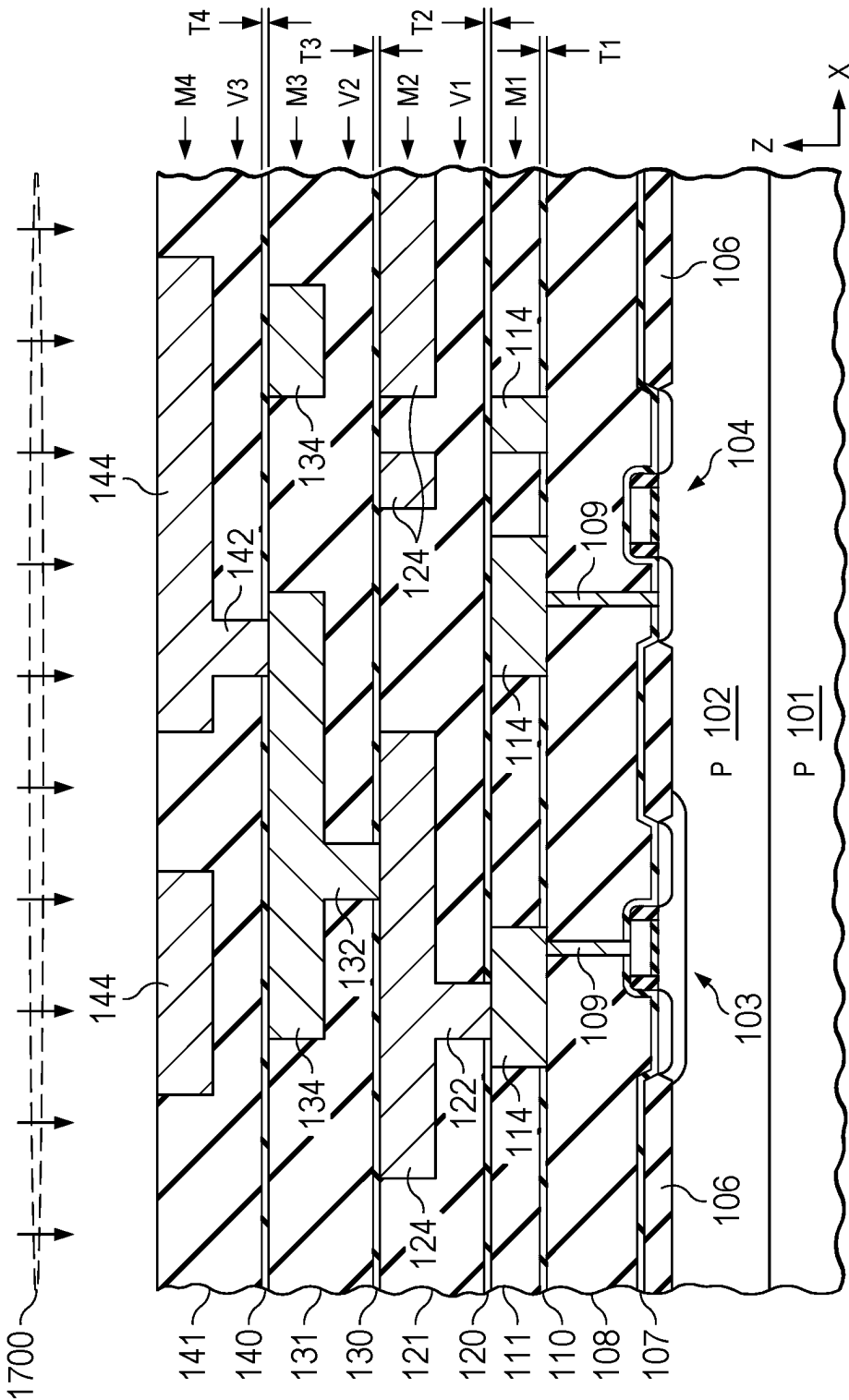


FIG. 17

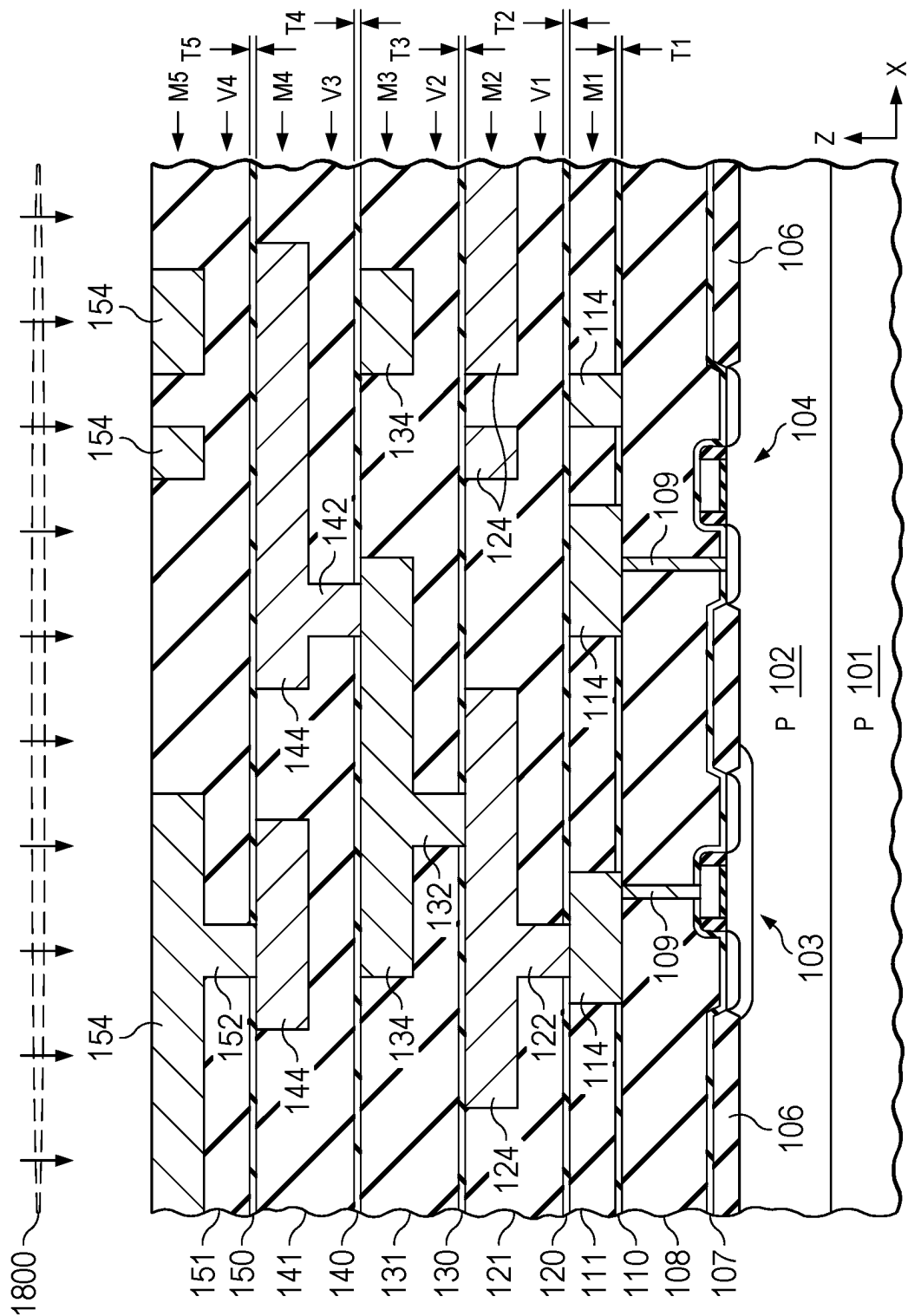
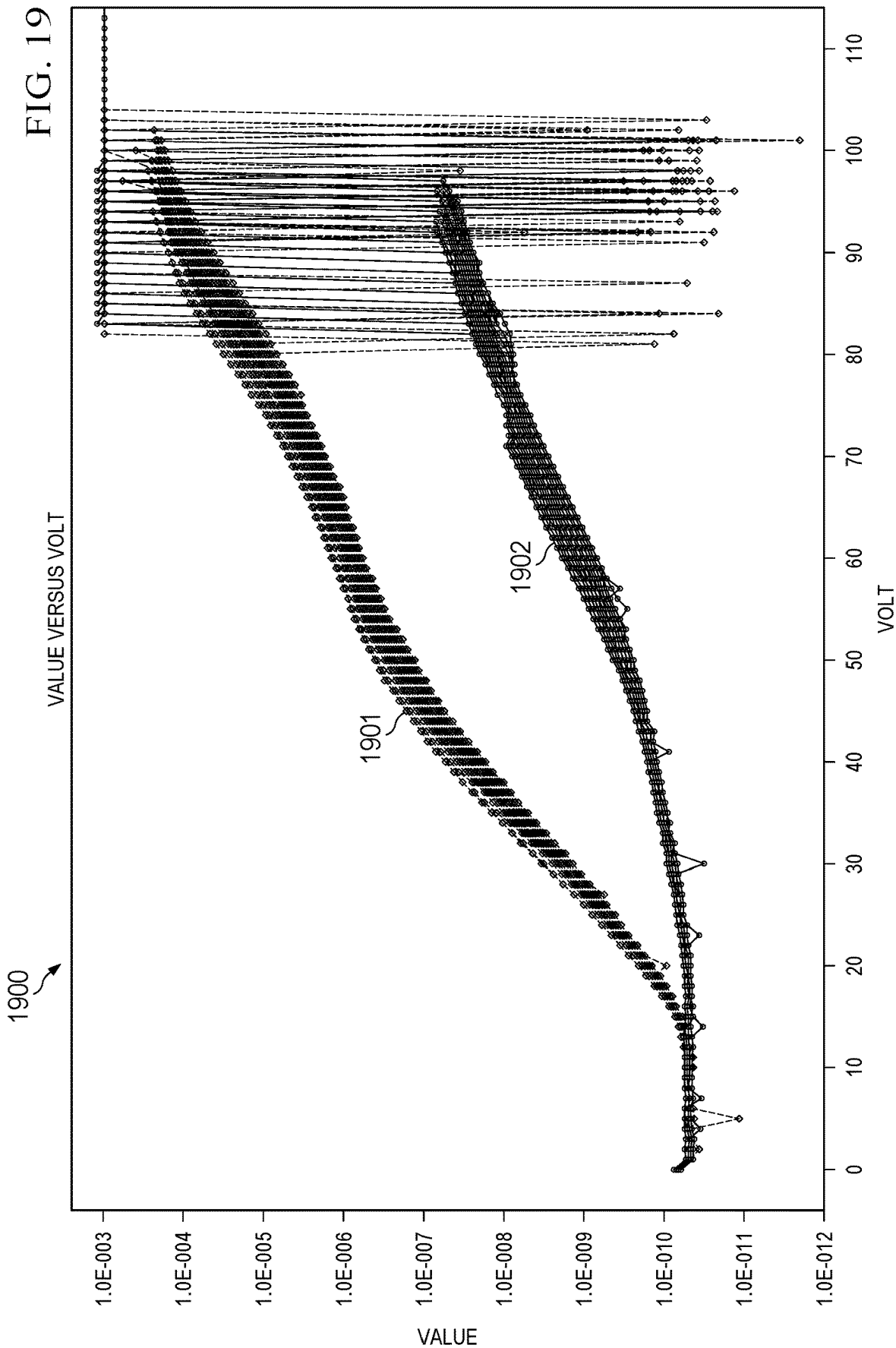
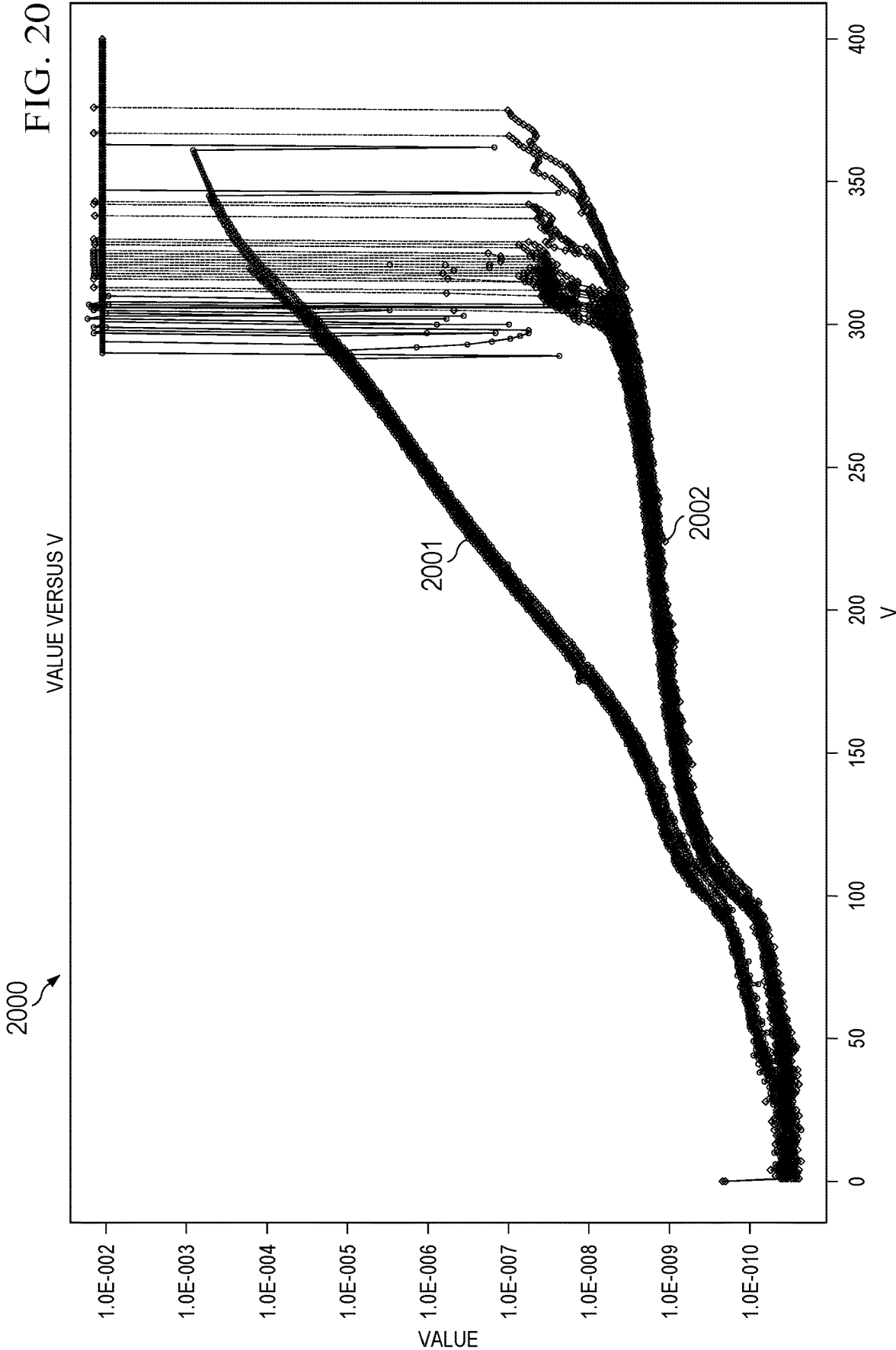


FIG. 18





1

DIELECTRIC SILICON NITRIDE BARRIER DEPOSITION PROCESS FOR IMPROVED METAL LEAKAGE AND ADHESION

BACKGROUND

Diffusion barriers are used in integrated circuits to mitigate diffusion of material from metal structures, such as copper metal lines or vias. In operation, current leakage can cause undesirable circuit performance, including intra-metal leakage across metal lines of a given metallization level and inter-metal current leakage between levels. Diffusion barriers can contribute to current leakage, and intra-metal current leakage can be more pronounced at high voltages.

SUMMARY

In one aspect, an electronic device includes a semiconductor die, a package structure, and a conductive lead. The package structure encloses a portion of the semiconductor die. The conductive lead is partially exposed outside the package structure and is electrically connected to a conductive feature of the semiconductor die. The semiconductor die has a multilevel metallization structure that includes stacked levels with respective dielectric layers and metal lines, and a diffusion barrier layer on one of the levels. The diffusion barrier layer contacts a side of the dielectric layer and a side of the metal line of the one of the levels. The diffusion barrier layer includes silicon nitride material having a first bond percentage ratio of ammonia to silicon nitride that is greater than a second bond percentage ratio of silicon hydride to silicon nitride.

In another aspect, an electronic device includes first and second metallization levels and a diffusion barrier layer in a semiconductor die. The first metallization level has a first dielectric layer and a first metal line. The diffusion barrier layer contacts a side of the first dielectric layer and a side of the first metal line. The diffusion barrier layer includes silicon nitride material that has a first bond percentage ratio of ammonia to silicon nitride of 6.5% or more and 10% or less, and a second bond percentage ratio of silicon hydride to silicon nitride of 1.5% or more and 1.9% or less. The second metallization level has a second dielectric layer on the diffusion barrier layer.

In a further aspect, a method of fabricating an electronic device includes forming a diffusion barrier layer on a side of a first dielectric layer and on a side of a first metal line, the diffusion barrier layer including silicon nitride material has a first bond percentage ratio of ammonia to silicon nitride of 6.5% or more and 10% or less, and a second bond percentage ratio of silicon hydride to silicon nitride of 1.5% or more and 1.9% or less, and forming a second dielectric layer on the diffusion barrier layer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a partial sectional side elevation view of a semiconductor die in an integrated circuit or other electronic device.

FIG. 1A is a top perspective view of the electronic device and semiconductor die of FIG. 1.

FIG. 2 is a flow diagram of a method of fabricating an electronic device.

FIGS. 3-18 are partial sectional side elevation views of a semiconductor wafer undergoing processing according to the method of FIG. 2.

2

FIGS. 19 and 20 show comparative current leakage performance data for intra-metal and inter-metal leakage.

DETAILED DESCRIPTION

In the drawings, like reference numerals refer to like elements throughout, and the various features are not necessarily drawn to scale. Also, the term “couple” or “couples” includes indirect or direct electrical or mechanical connection or combinations thereof. For example, if a first device couples to or is coupled with a second device, that connection may be through a direct electrical connection, or through an indirect electrical connection via one or more intervening devices and connections. One or more operational characteristics of various circuits, systems and/or components are hereinafter described in the context of functions which in some cases result from configuration and/or interconnection of various structures when circuitry is powered and operating. The example structures include layers or materials described as over or on another layer or material, which can be a layer or material directly on and contacting the other layer or material where other materials, such as impurities or artifacts or remnant materials from fabrication processing may be present between the layer or material and the other layer or material. Unless otherwise stated, “about,” “approximately,” or “substantially” preceded a value means ± 10 percent of the stated value.

Described electronic devices and methods provide a solution to mitigate inter-metal and intra-metal current leakage using a low leakage, low hydrogen diffusion barrier layer between metallization levels with reduced silicon-hydrogen bonds. In certain implementations, a short silane pretreatment before deposition of the low leakage, low hydrogen silicon nitride or silicon carbon nitride diffusion barrier layer materials provide further benefits of improved adhesion and electromigration reliability without significant adverse impact to metal resistance. In certain examples, the diffusion barrier layer mitigates diffusion of copper metal materials in a multilevel metallization structure during operation and/or production and also operates as an etch stop layer during device fabrication.

FIGS. 1 and 1A illustrate a packaged electronic device 100, such as an integrated circuit with a semiconductor die 180. As best shown in FIG. 1, the semiconductor die 180 includes a semiconductor substrate with a first semiconductor layer 101 and a semiconductor surface layer 102. The semiconductor layers 101 and 102 include semiconductor material (e.g., silicon, gallium nitride, etc.) doped by implantation and/or diffusion with dopants of a first conductivity type (e.g., p). In another implementation (not shown), one or both of the semiconductor layers 101 and/or 102 can be doped with dopants of an opposite second conductivity type (e.g., n). The electronic device 100 includes one or more electronic components such as transistors 103 and 104 that are formed on and/or in the semiconductor surface layer 102, along with isolation structures 106 (e.g., shallow trench isolation or STI structures, field oxide structures, etc.). The electronic device 100 in the illustrated example further includes a pre-metal dielectric (PMD) liner layer 107 that extends above (e.g., on) portions of the isolation structures 106 and the transistors 103 and 104. In one example, the PMD liner layer 107 is or includes silicon nitride (SiN) or stoichiometric variations thereof. A PMD dielectric layer 108 extends above (e.g., on) the PMD liner layer 107. In one example, the PMD dielectric layer 108 is or includes phosphosilicate glass (PSG) or borophosphosilicate glass (BPSG). Tungsten (W) or other conductive

contacts **109** extend through the PMD dielectric layer **108** to form electrical contacts or connections to terminals (e.g., gate, source, drain) of the transistors **103** and **104**.

A multilevel metallization interconnect structure extends above the PMD level. The metallization structure includes a first etch stop layer **110** that extends over the PMD dielectric layer **108** and forms an etch stop structure during fabrication of openings for first metal lines (e.g., labeled M1 in FIG. 1). In one example, the first etch stop layer **110** is or includes a low hydrogen, low leakage silicon nitride material (e.g., Si—N, such as SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H, such as NH₃) to silicon nitride that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). The first etch stop layer **110** has a first thickness T1, for example, approximately 600 Å. The first level of the metallization interconnect structure extends over the first etch stop layer **110** and includes a first inter-metal dielectric (IMD) dielectric layer **111** and metal lines **114**. In one example, the IMD dielectric layer **111** is or includes one of fluorosilicate glass (FSG), tetraethyl orthosilicate (TEOS), and silicon oxycarbide (e.g., SiCO). In the illustrated example, the first metal lines **114** are or include copper. In another example, a different conductive metal can be used, such as aluminum. Some of the illustrated first metal lines **114** in one example extend partially over, and directly contact, respective ones of the tungsten contacts **109** to form electrical connections thereto as shown in FIG. 1.

The multilevel metallization interconnect structure includes further levels arranged in a stacked configuration as shown in FIG. 1. A diffusion barrier layer **120** extends over the first metallization level and contacts an upper side of the dielectric layer **111** and upper sides of the first metal lines **114**. The diffusion barrier layer **120** has a thickness T2, for example, approximately 600 Å. The diffusion barrier layer **120** operates as an etch stop structure used during fabrication of openings for first vias (e.g., labeled V1 in FIG. 1). The diffusion barrier layer **120** is or includes a low hydrogen, low leakage silicon nitride material (e.g., SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H such as NH₃) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). The second level of the metallization interconnect structure extends over the diffusion barrier layer **120** and includes a second dielectric layer, referred to as an inter-level dielectric (ILD) layer **121** as well as the conductive metal first vias **122** (e.g., V1) and second metal lines **124** (e.g., labeled M2). In one example, the ILD dielectric layer **121** is or includes one of fluorosilicate glass (FSG), tetraethyl orthosilicate (TEOS), and silicon oxycarbide (e.g., SiCO). In the illustrated example, the first

vias **122** and the second metal lines **124** are or include copper. In another example, a different conductive metal is used, such as aluminum.

The third level in this example includes another diffusion barrier layer **130** that extends over the second metallization level and contacts an upper side of the dielectric layer **121** and upper sides of the second metal lines **124**. The diffusion barrier layer **130** has a thickness T3, for example, approximately 600 Å. The diffusion barrier layer **130** operates as an etch stop structure used during fabrication of openings for second vias **132** (e.g., labeled V2 in FIG. 1). The diffusion barrier layer **130** is or includes a low hydrogen, low leakage silicon nitride material (e.g., SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H such as NH₃) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). The third level of the metallization interconnect structure extends over the diffusion barrier layer **130** and includes a second ILD dielectric layer **131** as well as the conductive metal second vias **132** (e.g., labeled V2) and conductive third metal lines **134** (e.g., M3). In one example, the ILD dielectric layer **131** is or includes one of fluorosilicate glass (FSG), tetraethyl orthosilicate (TEOS), and silicon oxycarbide (e.g., SiCO). In the illustrated example, the second vias **132** and the third metal lines **134** are or include copper. In another example, a different conductive metal is used, such as aluminum.

A fourth metallization structure level includes a diffusion barrier layer **140** that extends over the third metallization level and contacts an upper side of the dielectric layer **131** and upper sides of the third metal lines **134**. The diffusion barrier layer **140** has a thickness T4, for example, approximately 600 Å. The diffusion barrier layer **140** operates as an etch stop structure used during fabrication of openings for third vias **142** (e.g., labeled V3). The diffusion barrier layer **140** is or includes a low hydrogen, low leakage silicon nitride material (e.g., SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H such as NH₃) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). The fourth level of the metallization interconnect structure extends over the diffusion barrier layer **140** and includes a third ILD dielectric layer **141** as well as the conductive metal third vias **142** (e.g., labeled V3) and conductive fourth metal lines **144** (e.g., M4). In one example, the ILD dielectric layer **141** is or includes one of fluorosilicate glass (FSG), tetraethyl orthosilicate (TEOS), and silicon oxycarbide (e.g., SiCO). In the illustrated example, the third vias **142** and the fourth metal lines **144** are or include copper. In another example, a different conductive metal is used, such as aluminum.

A fifth metallization structure level includes another diffusion barrier layer **150** that extends over the fourth metal-

lization level and contacts an upper side of the dielectric layer **141** and upper sides of the fourth metal lines **144**. The diffusion barrier layer **150** has a thickness **T5**, for example, approximately 600 Å. The diffusion barrier layer **150** operates as an etch stop structure used during fabrication of openings for fourth vias **152** (e.g., labeled **V4**). The diffusion barrier layer **150** is or includes a low hydrogen, low leakage silicon nitride material (e.g., SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H such as NH₃) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). The fifth level of the metallization interconnect structure extends over the diffusion barrier layer **150** and includes a fourth ILD dielectric layer **151** as well as the conductive metal fourth vias **152** (e.g., **V4**) and fifth metal lines **144** (e.g., **M5**). In one example, the ILD dielectric layer **151** is or includes one of fluorosilicate glass (FSG), tetraethyl orthosilicate (TEOS), and silicon oxycarbide (e.g., SiCO). In the illustrated example, the fourth vias **152** and fifth metal lines **154** are or include copper. In another example, a different conductive metal can be used, such as aluminum.

The electronic device **100** further includes another etch stop and diffusion barrier layer **160** with a thickness **T6** (e.g., approximately 600 Å) that extends over a portion of the ILD dielectric layer **151** and top sides of portions of some of the fifth metal lines **154**. The etch stop and diffusion barrier layer **160** is or includes a low hydrogen, low leakage silicon nitride material (e.g., SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H such as NH₃) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). A protective overcoat oxide layer **162** extends over the etch stop and diffusion barrier layer **160**, and a protective overcoat silicon oxynitride layer **164** (e.g., SiON) extends over the protective overcoat oxide layer **162**. A conductive metal cap structure **166** extends over and electrically contacts a portion of an upper side of one of the fifth metal lines **154**. In one example, the metal Structure **166** is or includes aluminum. In another example, a different conductive metal is used.

As further shown in FIG. 1A, the electronic device **100** as a molded package structure **168** and conductive gullwing leads **170**. The electronic device **100** is illustrated in FIGS. 1 and 1A in an example position in a three-dimensional space with respective first, second, and third mutually orthogonal directions X, Y, and Z. As best shown in FIG. 1A, the electronic device **100** includes opposite first and second sides **161** and **162** that extend along the second direction Y and are spaced apart from one another along the first direction X. The electronic device **100** also includes third and fourth sides **163** and **164** spaced apart from one another along the second direction Y, as well as a bottom side **165**, and a top side **166** spaced apart from the bottom side **165**

along the third direction Z. In the illustrated example, the bottom and top sides **165** and **166** are generally planar and extend in respective X-Y planes of the first and second directions X and Y. The electronic device **100** includes gullwing type conductive leads **170** along the first side **161**, as well as along the opposite second side **162**. The example gullwing leads **170** are or include conductive metal, such as aluminum or copper, and have internal portions enclosed by the molded package structure **168** as well as external portions partially exposed outside the package structure **168**. The semiconductor die **180** in this example has conductive features such as conductive bond pads that are electrically connected to respective ones of the conductive leads **170** via bond wires **182**.

The low leakage (LL), low hydrogen SiN or SiCN material of the layers **110**, **120**, **130**, **140**, **150**, and **160** have reduced (e.g., less) Si—H bonds compared with previously used silicon nitride at stop/diffusion barrier layers, and the reduced hydrogen bonds help to mitigate intra-metal and inter-metal leakage in operation. Moreover, the low leakage, low hydrogen diffusion barrier layers side affective protection against copper diffusion and also serve as at stop layers during fabrication of the electronic device **100**. In certain examples, one or more of the layers **110**, **120**, **130**, **140**, **150**, and/or **160** are formed by deposition processing following a short silane treatment to provide improved adhesion of the overlying subsequently deposited dielectric layer with limited increase in metal resistance. Unlike reduced hydrogen layer usage in preventing negative effects of hydrogen on PZT or other ferroelectric materials used in ferroelectric memory applications, the low hydrogen, low leakage silicon nitride or silicon carbon nitride material in the electronic device **100** advantageously mitigates both intra-metal and inter-metal current leakage in operation of the electronic device **100**, particularly in high voltage integrated circuits.

Referring also to FIGS. 2-18, FIG. 2 shows a method **204** fabricating an electronic device, and FIGS. 3-18 illustrate the electronic device **100** undergoing fabrication processing according to the method **200**. In FIGS. 3-18, the device is illustrated in wafer form prior to individual electronic devices being separated from a processed wafer and subsequently packaged to form a packaged electronic device **100** as shown in FIGS. 1-1A.

The method **200** includes front end of the line (FEOL) device fabrication at **202**, including processing steps (not shown) to fabricate the transistors **103** and **104** on or in the semiconductor surface layer **102** as shown in FIG. 3. The method **200** continues in FIG. 2 with pre-metal dielectric processing, including depositing a silicon nitride PMD liner layer at **211** (e.g., PMD liner layer **107** in FIG. 3), depositing the PMD dielectric layer at **212** (e.g., PMD dielectric layer **108**), and planarization such as chemical mechanical polishing (CMP) at **213** in FIG. 2. The PMD processing at **210** in this example also includes contact patterning and etching to form contact openings through the PMD dielectric layer **108** at **214**, deposition of tungsten or other conductive metal to fill the contact openings at **215**. FIG. 3 shows one example, in which a tungsten deposition process **300** is performed that deposits tungsten in the etched openings of the PMD dielectric layer **108**. The deposition process **300** forms further tungsten along the top side of the PMD dielectric layer **108** as shown in FIG. 3. The PMD processing at **210** in FIG. 2 further includes planarization such as CMP processing at **216**. FIG. 4 shows one example, in which a CMP process **400** is performed that planarizes the top side

of the processed wafer and exposes the top sides of the remaining tungsten contacts **109** and the PMD dielectric layer **108**.

The method **200** in FIG. **2** continues at **220** with first metallization level processing, including depositing a low hydrogen, low leakage etch stop layer at **222**. FIG. **5** shows one example, in which a plasma enhanced chemical vapor deposition (PECVD) process **500** is performed that deposits the first etch stop layer **110** on the upper sides of the PMD dielectric layer **108** and the tungsten contacts **109**. The first etch stop layer **110** is or includes a low hydrogen, low leakage silicon nitride material (e.g., SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H such as NH₃) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). The PECVD deposition process **500** in one example is performed at an ammonia NH₃ to silane SiH₄ flow ratio of 3.5 or more and 5 or less (e.g., approximately 4.3), and the PECVD deposition process **500** has a dinitrogen N₂ to silane SiH₄ flow ratio of 50 or more and 70 or less (e.g., approximately 60). In these or other examples, the PECVD deposition process **500** is performed at a pressure of 1.8 Torr or more and 2.6 Torr or less (e.g., approximately 2.25 Torr).

The first level processing at **220** in this example also includes depositing the inter-metal dielectric (IMD) dielectric layer at **223**. FIG. **6** shows one example, in which a deposition process **600** is performed that deposits the IMD dielectric layer **111** on the etch stop layer **110**. In one example, the IMD dielectric layer **111** is or includes one of fluorosilicate glass (FSG), tetraethyl orthosilicate (TEOS), and silicon oxycarbide (e.g., SiCO). The method **200** also includes patterning and etching at **224** to form the first metal level openings. FIG. **7** shows one example, in which an etch process **700** is performed using a mask **702** that forms opening **704** in the IMD dielectric layer **111**. At **225** and FIG. **2**, the method **200** also includes depositing copper or other conductive metal to fill the first metal openings. FIG. **8** shows one example, in which an electroplating or other copper deposition process **800** is performed that forms the first metal line features **114** in the previously etched openings of the IMD dielectric layer **111**. In this example, the deposition process **800** also forms copper over the top sides of the IMD dielectric layer **111**. At **226** in FIG. **2**, a planarization (e.g., CMP) is performed. FIG. **9** shows one example, in which a CMP process **900** is performed that planarizes the top side of the wafer to form the top sides of the first metal lines **114** and exposes the top sides of the IMD dielectric layer **111**.

The method **200** continues at **230** in FIG. **2** with processing to form the next metallization structure level. At **231** and **232**, a low hydrogen, low leakage silicon nitride or silicon carbon nitride diffusion barrier layer is formed over the planarized IMD level. At **231** in this example, the method **200** includes performing a silane pretreatment that exposes the side of the PMD dielectric layer **111** and the upper side of the first metal lines **114** to silane (e.g., SiH₄) prior to forming the diffusion barrier layer at **232**. FIG. **10** shows one example, in which a silane pretreatment process **1000** is performed that exposes the PMD dielectric layer **111** and the

first metal lines **114** to silane. In one example, the silane pretreatment process **1000** exposes the upper side of the first dielectric layer **111** and the upper side of the first metal line **114** to silane (e.g., SiH₄) for 0.5 second or more and 2.0 seconds or less. In another example, the silane pretreatment at **231** is omitted.

In the illustrated example, after the silane pretreatment at **231**, the method continues with depositing a low hydrogen, low leakage diffusion barrier layer at **232**. FIG. **11** shows one example, in which a plasma enhanced chemical vapor deposition (PECVD) process **1100** is performed that deposits the first diffusion barrier layer **120** on the upper sides of the IMD dielectric layer **111** and the first metal lines **114**. The first diffusion barrier layer **120** is or includes a low hydrogen, low leakage silicon nitride material (e.g., SiN) or silicon carbon nitride material (e.g., SiCN) having a first bond percentage ratio of ammonia (e.g., N—H such as NH₃) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). The PECVD deposition process **1100** in one example is performed at an ammonia NH₃ to silane SiH₄ flow ratio of 3.5 or more and 5 or less (e.g., approximately 4.3), and the PECVD deposition process **1100** has a dinitrogen N₂ to silane SiH₄ flow ratio of 50 or more and 70 or less (e.g., approximately 60). In these or other examples, the PECVD deposition process **1100** is performed at a pressure of 1.8 Torr or more and 2.6 Torr or less (e.g., approximately 2.25 Torr).

At **233**, the ILD dielectric layer **121** is deposited. FIG. **12** shows one example, in which a deposition process **1200** is performed that deposits the ILD dielectric layer **121** on the diffusion barrier layer **120**. In one example, the ILD dielectric layer **121** is or includes one of fluorosilicate glass (FSG), tetraethyl orthosilicate (TEOS), and silicon oxycarbide (e.g., SiCO). The method **200** also includes patterning and etching at **234** to form the first via and second metal line openings. FIG. **13** shows one example, in which a dual etch process **1300** is performed using a first mask (not shown) and a second mask **1302**. The process **1300** forms openings **1304** in the ILD dielectric layer **121**. At **235** and FIG. **2**, the method **200** includes depositing copper or other conductive metal to fill the first via and second metal line openings. FIG. **14** shows one example, in which an electroplating or other copper deposition process **1400** is performed that forms the first vias **122** and the second metal line features **124** in the previously etched openings of the IMD dielectric layer **121**. In this example, the deposition process **1400** also forms copper over the top sides of the ILD dielectric layer **121**. At **236** in FIG. **2**, a planarization (e.g., CMP) is performed. FIG. **15** shows one example, in which a CMP process **1500** is performed that planarizes the top side of the wafer to form the top sides of the second metal lines **124** and exposes the top sides of the ILD dielectric layer **121**.

In one example, the PECVD processes **500** and **1100** use a high frequency power level of approximately 1070 W at a frequency of approximately 13.56 MHz, and a low frequency power level of approximately 130 W at 400 kHz at a target spacing of approximately 0.6 inches in a Novellus Concept2 Sequel-6 tool, with a deposition rate of approximately 39 Å/S. One implementation of the processes **500**, **1100** uses an NH₃ flow rate of approximately 750 sccm, a

silane (e.g., SiH_4) flow rate of approximately 176 sccm, and an N_2 flow rate of approximately 14,000 sccm for a final thickness of approximately 600 Å and a SiN diffusion barrier material **110**, **120**, **130**, **140**, **150**, and/or **160** having a first bond percentage ratio of ammonia (e.g., N—H such as NH_3) to silicon nitride (e.g., Si—N) that is greater than a second bond percentage ratio of silicon hydride (e.g., SiH) to silicon nitride (e.g., SiN). The first bond percentage ratio may, for example, be 6.5% or more and 10% or less (e.g., approximately 8) and the second bond percentage ratio may be, for example, 1.5% or more and 1.9% or less (e.g., approximately 1.7). In this or another example, the first etch stop layer **110** may have a refractive index of 1.93 or more and 1.96 or less (e.g., approximately 1.93). In these or other examples, one or more of the etch stop/diffusion barrier layers **110**, **120**, **130**, **140**, **150**, and/or **160** have a third bond percentage ratio of NH to SiH of approximately 4.

A determination is made at **240** as to whether further metallization levels remain to be fabricated. If so (YES at **240**), the method **200** returns to repeat the processing at **230** for the next metallization interconnect level. FIG. **16** shows one example of the further processing at **230** to form the next ILD dielectric layer **131**, the second vias **132**, and the third metal lines **134**. Another level is then formed in the illustrated example at **230** in FIG. **2**, as shown in FIG. **17**, and FIG. **18** shows the wafer after a still further iteration at **230** to form the final interconnect level. Once all the desired levels are completed (NO at **240**), the method **200** proceeds at **242** with formation of the protective overcoat layers and bond pads or cap layers (e.g., **162**, **164**, and **166** in FIG. **1** above), individual dies are separated from the wafer at **224** (e.g., semiconductor die **180** in FIGS. **1** and **1A**), and the dies are packaged at **246** to provide packaged electronic devices (e.g., ICs) at **246**.

FIGS. **19** and **20** show respective graphs **1900** and **2000** that illustrate comparative current leakage performance data for intra-metal and inter-metal leakage. In the illustrated examples, lower NH_3/SiH_4 but higher N_2 flows in the deposition of the diffusion barrier layers **110**, **120**, **130**, **140**, **150**, and/or **160** produce SiN film with fewer SiH bond and more NH bonds. The graph **1900** in FIG. **19** shows a baseline current-voltage (I-V) curve **1901** and an I-V curve **1902** for example low hydrogen, low leakage silicon nitride diffusion barrier layers (e.g., **110**, **120**, **130**, **140**, **150**, and/or **160**) showing significant reduction in intra-metal leakage current as a function of applied voltage. The graph **2000** in FIG. **20** shows a baseline current-voltage (I-V) curve **2001** and an I-V curve **2002** for example low hydrogen, low leakage silicon nitride diffusion barrier layers (e.g., **110**, **120**, **130**, **140**, **150**, and/or **160**) showing significant reduction in combined inter-metal and intra-metal leakage current as a function of applied voltage with generally increased breakdown voltage (BV) levels. The graphs **1900** and **2000** demonstrate advantages and benefits facilitating increased operating voltage ratings and almost 100 percent leakage reduction for operating voltages of approximately 40 V. The use in some examples of the silane pretreatment (e.g., at **231** in FIG. **2**) enhances the benefits with respect to metallization structures. In this regard, conventional low hydrogen films were found to have lower leakage but poor interface properties with copper, and hence poor electromigration performance. Thus, while those films might serve as a hydrogen barrier for ferroelectric memories or other circuits, these types of films were believed to be unsuitable as effective diffusion barriers, particularly for use in contact with copper features of a multilevel metallization interconnect structure. Short silane and ammonia pretreat helps improve conditions

to improve interface adhesion and thus provide an effective copper diffusion barrier as well as a useful etch stop layer as evident through electrical monitor yields of via chain with wide metal links. Further benefits include enhanced pretreat with limited impact to resistance but good adhesion, low leakage due to stoichiometry of the film, lower free hydrogen for potential device stability improvement, and denser dielectric film to reduce copper hillocks.

Modifications are possible in the described examples, and other implementations are possible, within the scope of the claims.

What is claimed is:

1. A method of fabricating an electronic device, the method comprising:

forming a diffusion barrier layer on a side of a first dielectric layer and on a side of a metal line, the diffusion barrier layer including silicon nitride material, the diffusion barrier layer formed by a plasma enhanced chemical vapor deposition (PECVD) process that deposits the diffusion barrier layer on the side of the first dielectric layer and the side of the metal line, the PECVD process having an ammonia to silane flow ratio of 3.5 or more and 5 or less, and the PECVD process having a dinitrogen to silane flow ratio of 50 or more and 70 or less; and

forming a second dielectric layer on the diffusion barrier layer.

2. The method of claim 1, wherein the metal line includes copper.

3. The method of claim 1, further comprising:

performing a silane pretreatment that exposes the side of the first dielectric layer and the side of the metal line to silane before forming the diffusion barrier layer.

4. The method of claim 3, wherein the silane pretreatment exposes the side of the first dielectric layer and the side of the metal line to silane for 0.5 second or more and 2.0 seconds or less.

5. The method of claim 1, wherein the silicon nitride material has a first bond percentage ratio of ammonia to silicon nitride of 6.5% or more and 10% or less, and a second bond percentage ratio of silicon hydride to silicon nitride of 1.5% or more and 1.9% or less.

6. The method of claim 1, wherein the PECVD process is performed at a pressure of 1.8 Torr or more and 2.6 Torr or less.

7. The method of claim 1, wherein the diffusion barrier layer further includes carbon.

8. The method of claim 1, wherein the first and second dielectric layers comprise one of fluorosilicate glass, tetraethyl orthosilicate, and silicon oxycarbide.

9. A method of fabricating a semiconductor device, the method comprising:

forming a silicon nitride layer on a first dielectric layer, wherein the silicon nitride layer formed by a chemical vapor deposition (CVD) process including:

an ammonia to silane flow ratio of 3.5 or more and 5 or less; and

a dinitrogen to silane flow ratio of 50 or more and 70 or less; and

forming a second dielectric layer on the silicon nitride layer.

10. The method of claim 9, wherein the silicon nitride layer further forms on a metal structure.

11. The method of claim 10, wherein the metal structure includes copper or aluminum.

12. The method of claim 9, further comprising:
performing a pretreatment process to the first dielectric
layer, wherein the pretreatment process exposes the
first dielectric layer to silane prior to forming the
silicon nitride layer. 5
13. The method of claim 12, wherein the pretreatment
process lasts for 0.5 second or more and 2 seconds or less.
14. The method of claim 9, wherein the silicon nitride
layer includes a first bond percentage ratio of ammonia to
silicon nitride greater than a second bond percentage ratio of 10
silicon hydride to silicon nitride.
15. The method of claim 14, wherein:
the first bond percentage ratio of ammonia to silicon
nitride ranges between 6.5% and 10%; and
the second bond percentage ratio of silicon hydride to 15
silicon nitride ranges between 1.5% and 1.9%.
16. The method of claim 9, wherein the CVD process is
performed at a pressure of 1.8 Torr or more and 2.6 Torr or
less.
17. The method of claim 9, wherein the CVD process 20
includes:
an ammonia flow rate of approximately 750 sccm; and
a silane flow rate of approximately 176 sccm.
18. The method of claim 9, wherein the silicon nitride
layer has a refractive index of 1.93 or more and 1.96 or less. 25
19. The method of claim 9, wherein the silicon nitride
layer includes carbon.
20. The method of claim 9, wherein the first and second
dielectric layers comprise one of fluorosilicate glass, tetra-
ethyl orthosilicate, and silicon oxycarbide. 30

* * * * *